

Prompt 1 - Baseline geometry generation

Write a Python script to generate simple microfluidic channel centrelines for a spiral, a T-junction, and a Y-junction using parametric equations. Export the centrelines as STL-ready paths.

Prompt 2 - Hollow circular channels

Modify the script so that each centreline is swept with a circular profile to form hollow microfluidic channels with a specified outer diameter and wall thickness.

Prompt 3 - Watertight STL requirement

Ensure that the generated STL files are watertight and suitable for slicing. Avoid relying solely on slicer auto-repair.

Prompt 4 - Robust junction geometry

The T-junction and Y-junction STLs show non-manifold errors at the junctions. Redesign the geometry so junctions are robust and manifold, even after Boolean operations.

Prompt 5 - Precise T and Y Topology

Fix the T-junction so it is a true T (one horizontal main channel and one vertical branch from the centre). Fix the Y-junction so it has two symmetric branches at a specified angle (e.g., 45°).

Prompt 6 - Multi-format export and annotations

Extend the script to export: STL for fabrication, DXF with correct line widths, SVG images with clear annotations and ensure that each channel segment remains a separate polyline in DXF/SVG.

Prompt 7 - Logging and reproducibility

Add logging to the script: CSV file recording device type, generation time, and STL hash, Prompt log file that stores the text prompt used for SVG generation. This is for experimental reproducibility.

Prompt 8 - API-assisted SVG generation with fallback

Integrate the OpenAI API to generate SVG drawings of the microfluidic layouts when an API key is available. Include a timeout and a local SVG fallback if the API is unavailable.

Prompt 9 - Basic Description

Write a Python script for design and fabrication of hollow microfluidic devices for a spiral, a T-junction, and a Y-junction device. In this script, Python is used for geometry creation and fabrication ready STL export, and the API is used for SVG image generation. Using width=500 μ m, wall ratio=0.30, Spiral: ro=200 μ m, s=2000 μ m, loops=5. T and Y-Junction=2000 μ m (main length), v=1500 μ m (branch length), angle =45 degrees for Y-junction.

Prompt 10 - Final prompt that reproduces the full code

Write a single, complete Python script that generates Spiral, T-junction, and Y-junction microfluidic devices, uses circular hollow channels with parameters channel width $w = 500 \mu\text{m}$, wall ratio 0.30, spiral parameters $r_0 = 200 \mu\text{m}$, $s = 2000 \mu\text{m}$, loops = 5, T/Y parameters $h = 2000 \mu\text{m}$, $v = 1500 \mu\text{m}$, Y angle = 45° . Builds hollow channels using a robust CAD strategy: outer solid sweep – inner solid sweep (OpenSCAD Boolean difference). Produces watertight STL files suitable for slicing without auto-repair. Exports DXF and SVG drawings where each channel is a separate polyline. Uses GPT-5 via the OpenAI API to generate SVGs when an API key is available, with a local SVG fallback. Adds clear geometry annotations to DXF/SVG outputs. Logs device runs to a CSV file and stores SVG prompts in a prompt log. Includes OpenSCAD and trimesh fallbacks, robust error handling, and reproducibility metadata. Output directory should be created automatically, and the script should run end-to-end when executed.